Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/31/2006
PI Name:	Theofanous, Theofanis G. Ph.D.		
Project Title:	The Pool Boiling Crisis From Flat Plates: Mechanism(s) and Enhancement		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Multiphase flow and heat transfer		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	93106	**Congressional District:**	23
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	96-HEDS-01
Start Date:	04/01/1998	**End Date:**	12/31/2004
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	Note - date changed from 12/31/2001		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Dinh, Truc-Nam (University of California Santa Barbara)		
Grant/Contract No.:	None		
Performance Goal No.:	2.6		
Performance Goal Text:	Advance the state of knowledge sufficiently to allow development of reliable and efficient heat transfer technology for space and extraterrestrial operations.		

Task Description:	Phenomena of boiling heat transfer under high heat flux and boiling crisis on horizontal flat plates are investigated. An integrated experimental, theoretical and numerical approach is used. Initially, the task was focused on quantification of hydrodynamic parameters, to examine the widely accepted hydrodynamic concept of boiling crisis. Experimental work performed in the course of this study using advanced diagnostic techniques pointed to the controlling role of heater surface nanomorphology and chemistry. Based on this information, the research was re-focused toward characterization of the effect of heater surface on high heat flux boiling and on mechanisms of boiling crisis. The objective is to develop and demonstrate a basic understanding of the Boiling Crisis phenomenon and establish a rational basis for its quantification (in terms of critical heat flux values) and control. Understanding the boiling phenomena has tremendous impact on electronic cooling and the nuclear, oil, and electric power industries since these industries rely heavily on transferring heat from one location to another through the latent heat of vaporization. Space-based processing, power systems, transfer of fluids and heat, and environmental control also require an understanding of boiling phenomena and of burnout as a limiting condition.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 10/17/2007)